# A novel technology to measure borehole trajectory using an electronic borehole compass

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## "An indigenous device to determine the borehole trajectory"



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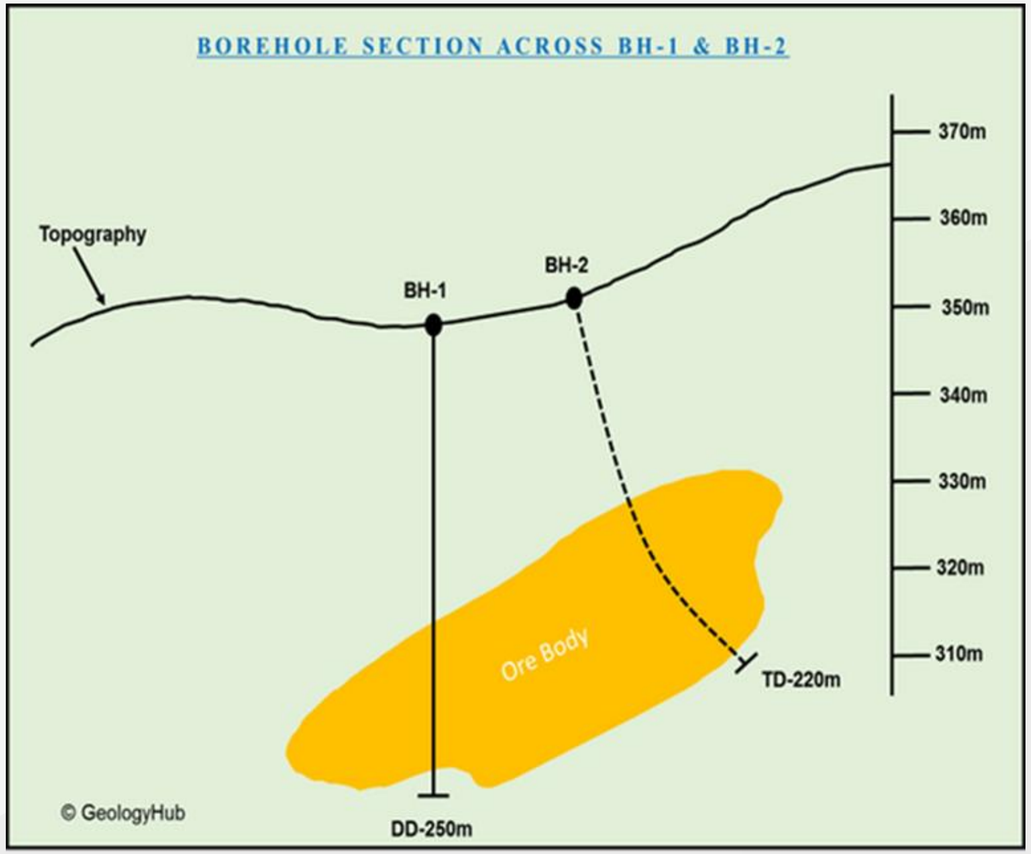
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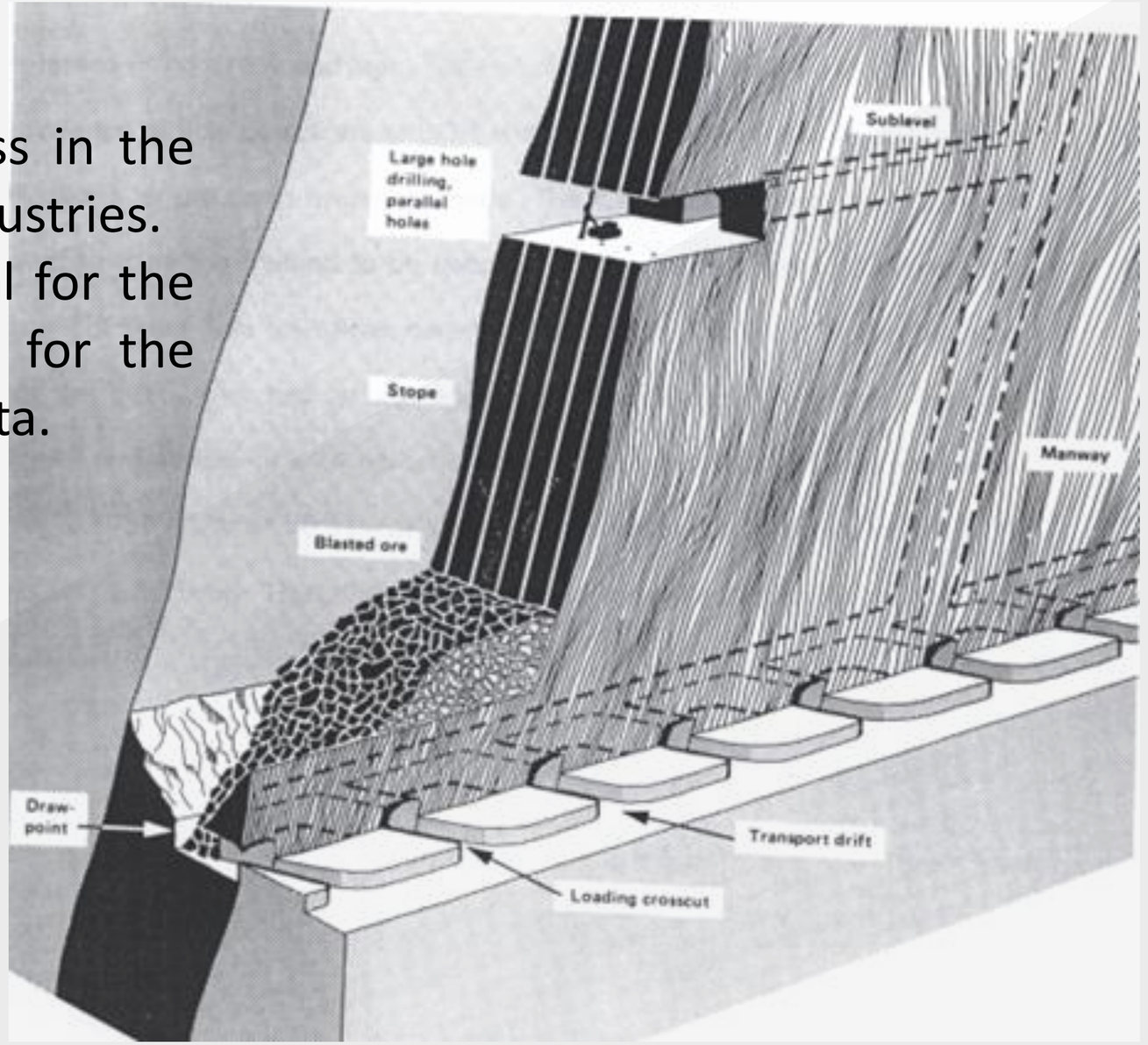
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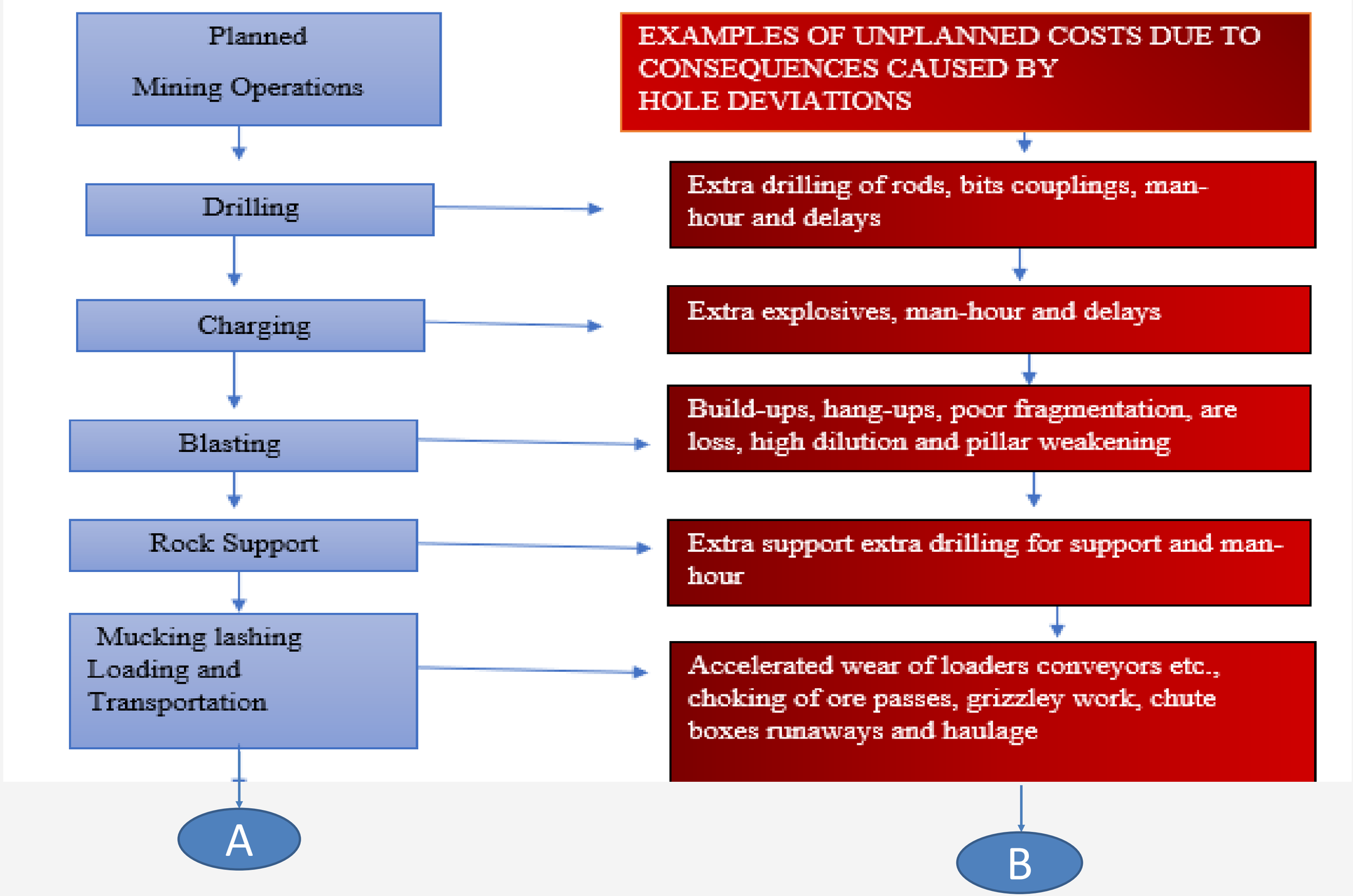
Associate Prof. Dept. of Mining Engg, IIT (ISM) Dhanabd

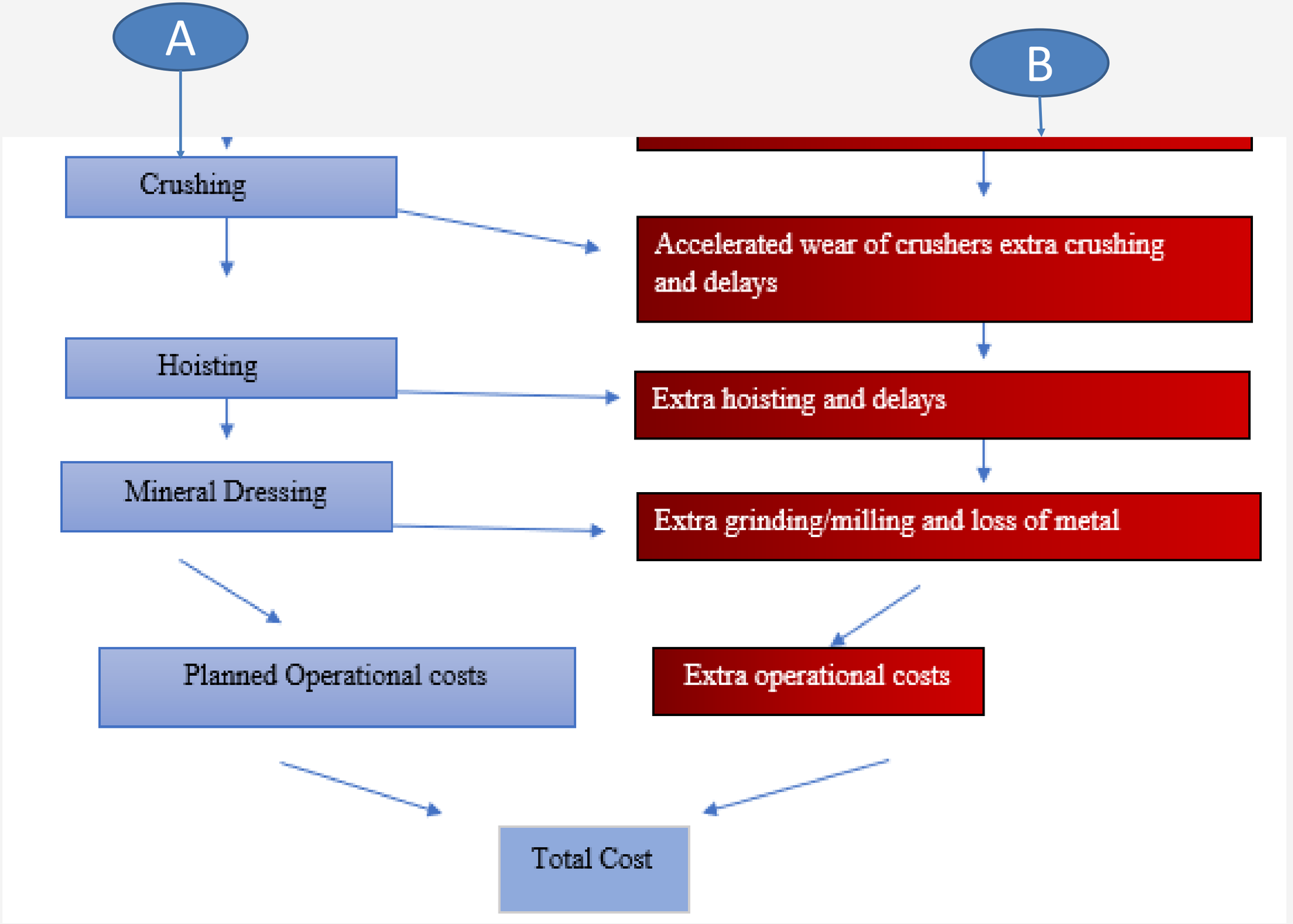
# Introduction

- Borehole drilling is an inevitable process in the field of mining engineering and allied industries.
- Correctness in borehole drilling is crucial for the accurate estimation of ore bodies and for the optimum fragmentation of host rock strata.




Planned
Mining Operations
Drilling
Charging
Blasting
Rock Support
Mucking lashing
Loading and
Transportation
A
EXAMPLES OF UNPLANNED COSTS DUE TO CONSEQUENCES CAUSED BY HOLE DEVIATIONS
Extra drilling of rods, bits couplings, man-hour and delays
Extra explosives, man-hour and delays
Build-ups, hang-ups, poor fragmentation, are loss, high dilution and pillar weakening
Extra support extra drilling for support and man-hour
Accelerated wear of loaders conveyors etc., choking of ore passes, grizzley work, chute boxes runaways and haulage
B

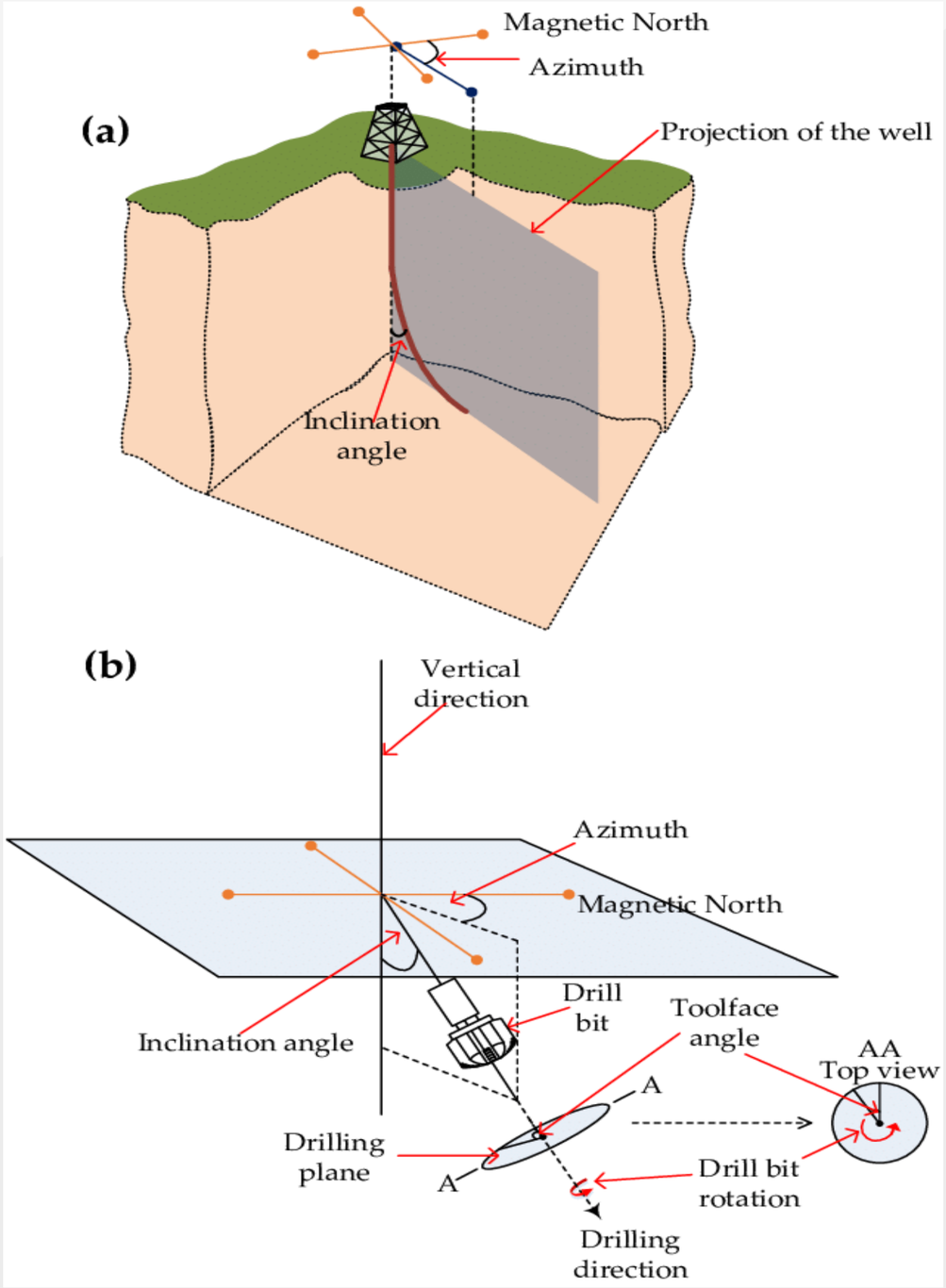

A
B
Crushing
Accelerated wear of crushers extra crushing and delays
Hoisting
Extra hoisting and delays
Mineral Dressing
Extra grinding/milling and loss of metal
Planned Operational costs
Extra operational costs
Total Cost

# Fundamental of Technology

- To fully determine the orientation of the borehole, two parameters have to be known: the angle between the borehole axis and true vertical (inclination) and the angle of its departure from magnetic North (Azimuth).

- To describe the orientation of a line in space, two pieces of information are required - trend and plunge. The trend is defined as the direction of a line on a horizontal plane.

- We use the Azimuth of a line to define its trend. Azimuth is known as a horizontal angle (between 0 and 360°) measured clockwise from true North



# Electronic Circuitry

- An accelerometer sensor (fixed inside the compass module) is used to sense the yaw, pitch, and roll about its axis, sending it to the Microcontroller.
- A program to convert roll, pitch, and yaw to trend and plunge (Spatial coordinate transformation) is written in Microcontroller, parsing every incoming accelerometer sensor data to trend and plunge.
- Also, a Magnetometer sensor is used to give the precise azimuthal orientation of the module.
- The Microcontroller receives the data from both sensors, making transformations, and then logs (time, trend, plunge, and Azimuth) to the attached SD card every 1 second.

# *Area of application*

**Exploratory Drilling**

Recording, analyzing and keeping a detailed record of drilling deviation parameters by plotting their graphs

**Long hole Drilling**

Identification of blast hole deviation in ring drilling during long hole stopinging metal mines

**Fracture Orientation Measurement**

In hydraulic fracturing method-for estimation of the in-situ stress direction

# TECHNOLOGY AND MARKET GAP

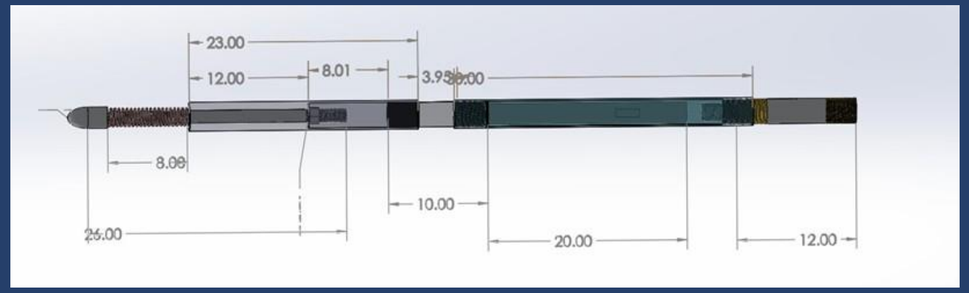
## High cost

The currently available Electronic Borehole Compass (EBC) involves huge cost investment in order to put it to use in the mining industry

## Tedious and inefficient to measure borehole deviation in real-time

Most of the conventional devices currently available don’t have the ability to continuously monitor the accuracy in drilling boreholes with correct orientation with the degree of accuracy that our instrument provides.

# Design and Machining of the Mechanical Part




Solid models of electronics borehole compass along with its dimensions






Turning process of Aluminum Rod in Workshop



Shock being absorber by spring provided at the front of device



# CALIBRATION OF THE EBC

Images show calibration of electronic Borehole compass module in the Surveying lab, IIT(ISM) DHANBAD. The developed electronics compass has been calibrated using a total station in the Surveying lab.



# 1st FIELD TESTING

The compass was successfully tested in the Moonidih Mines (at BCCL)



# 2nd FIELD TESTING

The compass was successfully tested in the Churcha Mines RO(SECL)






**IIT (ISM) team with Churcha Mine (RO) officials, SECL**

# KEY PRODUCTS AND SERVICES

## Functionalities analysis with other competitors:

| Functions | Aimil | BlastGyro | Our Device |
|---|---|---|---|
| SD Card Storage | ✓ | ✗ | ✓ |
| Battery type | ✓ | ✓ | ✓ |
| Azimuth | ✓ | ✓ | ✓ |
| Field control unit ( excel unit ) | ✓ | ✓ | ✓ |
| Data transfer (USB) | ✓ | ✓ | ✓ |
| Operation system ( windows mobile 6.1 ) | ✓ | ✓ | ✗ Notepad |
| Survey software | ✓ | ✓ | ✗ in process |

# THANK YOU